Michelle Ntampaka, Ph.D.

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Research Interests

Machine learning is emerging as a powerful tool for astronomical discovery and science operations. My research focuses on developing trustworthy new techniques for interpreting cosmological observations, with a focus on using machine learning to make new physical discoveries. My first career as a teacher provided me with broad pedagogical training and team management experience.

Education

2024 ∘ Organizational Leadership Executive Certificate ∘ JHU Carey Business School ∘ Baltimore, MD
MBA-level Coursework in topics including project management, organizational change, risk analysis, and decision-making.

2017 ∘ Ph.D. ∘ Physics ∘ Carnegie Mellon University (CMU) ∘ Pittsburgh, PA
Advisor: Hy Trac
Thesis: Cosmology With Galaxy Cluster Dynamics Using Machine Learning & Forward Modeling

2010 ∘ M.S. ∘ Physics ∘ CMU
Earned while holding full-time employment.

2004 ∘ B.S. ∘ Physics & Science Secondary Education ∘ Grove City College (GCC) ∘ Grove City, PA

2003 ∘ B.S. ∘ Applied Physics, Minor in Mathematics ∘ GCC

Professional Experience

2020-present ∘ Data Science Mission Office ∘ Space Telescope Science Institute ∘ Baltimore, MD

Role:	Dates:
Deputy Mission Head	June 2022 - present
Acting Mission Head	February 2026 - July 2026
Acting Mission Head	February 2023 - August 2023
Acting Deputy Mission Head	January 2022 - May 2022
Data Scientist	September 2020 - December 2021

2020-present ∘ Tenure-Track Astronomer ∘ STScI ∘ Baltimore, MD

Role:	Dates:
Tenured Associate Astronomer	March 2026 - present
Associate Astronomer w/o tenure	February 2024 - March 2026
Assistant Astronomer	September 2020 - February 2024

2020-present ∘ Associate Research Scientist ∘ Department of Physics and Astronomy
Johns Hopkins University (JHU) ∘ Baltimore, MD

2017-2020 ∘ Postdoctoral Fellow ∘ Data Science Initiative ∘ Harvard University ∘ Cambridge, MA
Mentors: Daniel Eisenstein & Alexey Vikhlinin

2012-2017 ∘ Research Assistant and Ph.D. Student ∘ CMU

2005-2012 ∘ Special Lecturer & Laboratory Technician ∘ CMU

2004-2005 ∘ Physics Teacher ∘ Baldwin High School ∘ Pittsburgh, PA

2003-2004 ∘ Part-Time Middle School Science Teacher ∘ Grove City Academy ∘ Grove City, PA

2003-2004 ∘ Secondary Education Training & Student Teaching ∘ GCC

Highlights

- Author of influential white papers on AI and scientific practice.
- Internationally invited speaker on machine learning, cosmology, and AI for science.
- P.I. of over $1.5 million in competitive research funding, including a NASA Astrophysics Theory Program Grant and an NSF Astronomy & Astrophysics Research Grant.
- Key strategic leader for Institute-wide AI initiatives supporting the Hubble, JWST, and Roman Space Telescopes.

Fellowships, Awards, & Grants

P.I. ∘ NSF Astronomy and Astrophysics Research Grants (AAG) ∘ $457,000 ∘ 2023-2026
"A Modern Toolkit to Enhance the Scientific Productivity of Optical Survey Data"
Awarded as a part of the NSF Computational and Data-Enabled Science and Engineering (CDS&E) Program

co-I. ∘ ESA Faculty Research Support ∘ €150,000 (≈ $160,000 at time of award) ∘ 2022-2023
"Exploring the Evolution of Galaxy Morphology in Different Environments with Zoobot"
P.I.: Bruno Merín, ESA

co-I. ∘ *Chandra* Theory Grant ∘ Amount TBD (≈ $100,000) ∘ 2022-2023
"A Photon Point Cloud Machine Learning Approach for Galaxy Cluster X-ray Mass Estimation"
P.I.: Daisuke Nagai, Yale University

P.I. ∘ NASA Astrophysics Theory Program (ATP) ∘ $543,000 ∘ 2022-2025
"Precision Cluster Cosmology with Interpretable Machine Learning"

P.I. ∘ NASA Compute Resources ∘ 900,000 standard billing units ∘ valued at $396,000 ∘ 2022-2025
Note: In addition to the grant listed in the item above, the NASA-funded research program "Precision Cluster Cosmology with Interpretable Machine Learning" was also awarded this compute allocation at the NASA Advanced Supercomputing (NAS) Facility.

P.I. ∘ STScI Director's Discretionary Research Funds (DDRF) ∘ $56,000 ∘ 2020-2021
"Enhancing Ground-Based Galaxy Observations with Machine Learning"

P.I. ∘ Multiple Mirror Telescope (MMT) Binospec Observing Time ∘ $23,000 ∘ 2020†

P.I. ∘ eROSITA External Collaborator ∘ 2020

P.I. ∘ Multiple Mirror Telescope (MMT) Binospec Observing Time ∘ $23,000 ∘ 2019

P.I. ∘ Pittsburgh Supercomputing Center Start-up Grant ∘ $800 ∘ 2018

McWilliams Presidential Fellowship ∘ Mellon College of Science ∘ CMU ∘ $70,000 ∘ 2016
- ∘ Full scholarship and stipend for a "record of distinction and the potential for further excellence."

Science Communication Fellowship ∘ inaugural class ∘ Carnegie Science Center ∘ 2015

Astronomy Ambassador ∘ inaugural class ∘ AAS ∘ 2013

M. Hildred Blewett Fellow ∘ American Physical Society ∘ $45,000 ∘ 2012
- ∘ One of two recipients of this national fellowship for women physicists returning to careers in physics.

Research in the News

What's your lab's archetype? The answer could inform how you use AI
by Jenna Ahart, 2026 Nature Career News.

Hey Siri, How Much Does This Galaxy Cluster Weigh?
by Amy Pavlak Laird, 2022 Mellon College of Science News.

Faced with a Data Deluge, Astronomers Turn to Automation
by Anil Ananthaswamy, 2019 Scientific American.

Refereed Publications

NOTE: Citations for interdisciplinary research are not cataloged well in any single tracking platform, and ADS frequently omits citations from the computer science literature.
See ADS for a current publication list with astronomy citations.
See Google Scholar for a current record of interdisciplinary citations.
See ORCiD (ID: 0000-0002-0144-387X) for a current publication record.

Mentees who were primarily advised on the research publications below are bolded and underlined.
Mentees* who received significant support (30-50%) for the research publications below are marked with an asterisk.
Early Career Scientists† who were mostly mentored by another senior scientist, but who received significant financial support under my PIship for the research performed are marked with a dagger.

29. LoVoCCS. III. Third Generation Pipeline & The Hercules Supercluster
A.M. Englert, S. Fu, I. Dell'Antonio, M.H. Abdullah, S. Abdulshafy, E. Aljamal, W.K. Black, N. Chidester, D. Clowe, M.C. Cooper, M. Donahue, Z. Escalante, A. Evrard, E. Habjan, S. Helhoski, B. Liu, J. McCleary, H. Miyatake, M. Montes, P. Natarajan, J. Nelson, **Michelle Ntampaka**, E. Pierpaoli, M. Postman, R. Shinde, J. Sohn, F. Tanaka, D. Turner, K. Umetsu, Y. Utsumi, R. Wang, G. Wilson
2026, ApJS, *submitted*.

28. Estimating the triaxiality of massive clusters from 2D observables in MillenniumTNG with machine learning
Ana Maria Delgado, **Michelle Ntampaka**, S. Bose, F. Ferlito, B. Hadzhiyska, L. Hernquist, J. Soltis, J.F. Wu, M. Yunus, J. ZuHone
2026, ApJ, *in press*.

27. The Importance of Being Adaptable: An Exploration of the Power and Limitations of Domain Adaptation for Simulation-Based Inference with Galaxy Clusters
Michelle Ntampaka, A. Ćiprijanović, A. Delgado, J. Soltis, J. Wu, M. Yunus, J. ZuHone.
2025 arXiv, *submitted to ApJ*.

26. Galaxy Cluster Characterization with Machine Learning Techniques
M. Sadikov, J. Hlavacek-Larrondo, L. Perreault Levasseur, C.L. Rhea, M. McDonald, **Michelle Ntampaka**, J. ZuHone.
2025 ApJ, 986, 24.

25. A Multi-Wavelength Technique for Estimating Galaxy Cluster Mass Accretion Rates
John Soltis, **Michelle Ntampaka**, B. Diemer, J. ZuHone, S. Bose, A.M. Delgado, B. Hadzhiyska,

C. Hernandez-Aguayo, D. Nagai, H. Trac.
2025 ApJ, 985, 212.

24. Reconstructing Galaxy Cluster Mass Maps using Score-based Generative Modeling
Alan Hsu[†], M. Ho, J. Lin, C. Markey, **Michelle Ntampaka**, H. Trac, B. Póczos.
2025 Open Journal of Astrophysics, 8, 92.

23. LoVoCCS – II. Weak Lensing Mass Distributions, Red-Sequence Galaxy Distributions, and Their Alignment with the Brightest Cluster Galaxy in 58 Nearby X-ray-Luminous Galaxy Clusters
S. Fu, I. Dell'Antonio, Z. Escalante, J. Nelson, A. Englert, S. Helhoski, R. Shinde, J. Brockland, P. LaDuca, C. Larkin, L. Paris, S. Weiner, W.K. Black, R.-R. Chary, D. Clowe, M.C. Cooper, M. Donahue, A. Evrard, M. Lacy, T. Lauer, B. Liu, J. McCleary, M. Meneghetti, H. Miyatake, M. Montes, P. Natarajan, **Michelle Ntampaka**, E. Pierpaoli, M. Postman, J. Sohn, D. Turner, K. Umetsu, Y. Utsumi, G. Wilson.
2024 ApJ, 974, 69.

22. Designing an Evaluation Framework for Large Language Models in Astronomy Research
J.F. Wu, A. Hyk, K. McCormick, C. Ye, S. Astarita, E. Baral, J. Ciuca, J. Cranney, A. Field, K. Iyer, P. Koehn, J. Kotler, S. Kruk, **Michelle Ntampaka**, C. O'Neill, J.E.G. Peek, S. Sharma, M. Yunus.
2024 arXiv: 2405.20389

21. Painting baryons onto N-body simulations of galaxy clusters with image-to-image deep learning
Urmila Chadayammuri, **Michelle Ntampaka**, J. ZuHone, Á. Bogdán, and R.P. Kraft.
2023 MNRAS, 526, 2812-2829.

20. Benchmarks and Explanations for Deep Learning Estimates of X-ray Galaxy Cluster Masses
M. Ho, J. Soltis, A. Farahi, D. Nagai, A. Evrard, and **Michelle Ntampaka**.
2023 MNRAS, 524, 3289-3302.

19. Predicting the impact of feedback on matter clustering with machine learning in CAMELS
A.M. Delgado, D. Angles-Alcazar, L. Thiele, **Michelle Ntampaka**, S. Pandey, K. Lehman, R. Somerville, S. Genel, F. Villaescusa-Navarro.
2023 MNRAS, 526, 4.

18. A Machine Learning Approach to Enhancing *eROSITA* Observations
John Soltis, **Michelle Ntampaka**, J. Wu, J. ZuHone, A. Evrard, A. Farahi, M. Ho, D. Nagai.
2023 Astrophysical Journal, 940, 1.

17. Deep Learning and the Dynamical Mass of the Coma Cluster
Matthew Ho[*], **Michelle Ntampaka**, M.M. Rau, M. Chen, A. Lansberry, F. Ruehele, H. Trac.
2022 Nature Astronomy, 6.

16. LoVoCCS - I. Survey Introduction, Data Processing Pipeline, and Early Science Results
S. Fu, I. Dell'Antonio, R. Chary, D. Clowe, M. Cooper, M. Donahue, A. Evrard, M. Lacy, T. Lauer, B. Liu, J. McCleary, M. Meneghetti, H. Miyatake, M. Montes, P. Natarajan, **Michelle Ntampaka**,

E. Pierpaoli, M. Postman, J. Sohn, K. Umetsu, Y. Utsumi, and G. Wilson.
2022 Astrophysical Journal, 933, 84.

15. The Importance of Being Interpretable: Toward An Understandable Machine Learning Encoder for Galaxy Cluster Cosmology
Michelle Ntampaka and A. Vikhlinin.
2022 Astrophysical Journal, 926, 45.

14. Building Trustworthy Machine Learning Models for Astronomy
Michelle Ntampaka, M. Ho, and B. Nord.
Astronomical Data Analysis Software and Systems (ADASS) XXXI Proceedings.
NOTE: Computer Science conference papers are peer-reviewed.

13. Emulating Sunyaev-Zeldovich Images of Galaxy Clusters using Auto-Encoders
T. Rothschild, D. Nagai, H. Aung, S. Green, **Michelle Ntampaka**, J. ZuHone.
2022 Monthly Notices of the Royal Astronomical Society, 513, 333.

12. SuperRAENN: A Semi-supervised Photometric Classification Pipeline for Spectroscopically-Classified Pan-STARRS1 Supernovae
V. Ashley Villar, G. Hosseinzadeh, E. Berger, **Michelle Ntampaka**, D.O. Jones, P. Challis, R. Chornock, M.R. Drout, R.J. Foley, R.P. Kirshner, R. Lunnan, R. Margutti, D. Milisavljevic, N. Sanders, Y. Pan, A. Rest, D.M. Scolnic, E. Magnier, N. Metcalfe, R. Wainscoat, C. Waters.
2020 Astrophysical Journal, 905, 94.

11. A Deep Learning View of the Census of Galaxy Clusters in IllustrisTNG
Y. Su, Y. Zhang, G. Liang, J.A. ZuHone, D. Barnes, N.B. Jacobs, **Michelle Ntampaka**, W.R. Forman, P.E.J. Nulsen, R.P. Kraft, C. Jones.
2020 Monthly Notices of the Royal Astronomical Society, 498, 5620.

10. A Hybrid Deep Learning Approach to Cosmological Constraints From Galaxy Redshift Surveys
Michelle Ntampaka, D.J. Eisenstein, S. Yuan, L. Garrison.
2020 Astrophysical Journal, 889, 151.

9. A Robust and Efficient Deep Learning Method for Dynamical Mass Measurements of Galaxy Clusters
M. Ho, M.M. Rau, **Michelle Ntampaka**, A. Farahi, H. Trac, B. Póczos.
2019 Astrophysical Journal, 887, 25.

8. Using X-ray Morphological Parameters to Strengthen Galaxy Cluster Mass Estimates via Machine Learning
Sheridan B. Green, **Michelle Ntampaka**, D. Nagai, L. Lovisari, K. Dolag, D. Eckert, J.A. ZuHone.
2019 Astrophysical Journal, 884, 33.

7. Cluster Cosmology With the Velocity Distribution Function of the HeCS-SZ Sample
Michelle Ntampaka, K. Rines, H. Trac.
2019 Astrophysical Journal, 880, 154.

6. Machine Learning Applied to the Reionization History of the Universe
   P. La Plante, **Michelle Ntampaka**.
   2019 Astrophysical Journal, 880, 110.

5. A Deep Learning Approach to Galaxy Cluster X-ray Masses
   Michelle Ntampaka, J. Zuhone, D. Eisenstein, D. Nagai, A. Vikhlinin, L. Hernquist, F. Marinacci, D. Nelson, R. Pakmor, A. Pillepich, P. Torrey, M. Vogelsberger.
   2019 Astrophysical Journal, 876, 82.

4. The Velocity Distribution Function of Galaxy Clusters as a Cosmological Probe
   Michelle Ntampaka, H. Trac, J. Cisewski, L.C. Price.
   2017 Astrophysical Journal, 835, 106.

3. Dynamical Mass Measurements of Contaminated Galaxy Clusters Using Machine Learning
   Michelle Ntampaka, H. Trac, D.J. Sutherland, S. Fromenteau, B. Póczos, J. Schneider.
   2016 Astrophysical Journal, 831, 135.

2. A Machine Learning Approach for Dynamical Mass Measurements of Galaxy Clusters
   Michelle Ntampaka, H. Trac, D.J. Sutherland, N. Battaglia, B. Póczos, J. Schneider.
   2015 Astrophysical Journal, 803, 50.

1. A First Look at Creating Mock Catalogs With Machine Learning Techniques
   X. Xu, S. Ho, H. Trac, J. Schneider, B. Póczos, **Michelle Ntampaka**.
   2013 Astrophysical Journal, 772, 147.

Other Publications

6. How to Craft the Right Language AI Policy For Your Research Group (Some Assembly Required)
   Michelle Ntampaka, S. Burke-Spolaor, I. Ciucă, A.M. Delgado, K.G. Iyer, K. Lockhart, A. Plunkett, M. López-Morales, A.A. Mahabal, S.E. Mullally, R. Raj, J. Reerink, J. Smith, J.S. Speagle, J. Soltis, J.F. Wu, M. Yunus
   2026, arXiv:2607.20836.

5. Constructing Impactful Machine Learning Research for Astronomy: Best Practices for Researchers and Referees
   co-first-authorship: Daniela Huppenkothen and Michelle Ntampaka, M. Ho, M. Fouesneau, B. Nord, J.E.G. Peek, M. Walmsley, and J.F. Wu.
   2023 arXiv: 2310.12528
   NOTE: this article was in the top 100 most-discussed astronomy papers of 2023 (according to the journal club website Benty-Fields).

4. A Referee Primer
   Michelle Ntampaka, A. Bonaca, S. Bose, D.J. Eisenstein, B. Hadzhiyska, C. Mason, D. Nagai, and J.S. Speagle.
   2022 Bulletin of the AAS, 54.
   NOTE: at my request, this manuscript underwent peer review by a member of the AAS editorial staff. In the first month after publication, this was downloaded more than 1200 times (according to ADS) and was in the top 35 most-discussed astronomy papers of 2022 (according to the journal club website Benty-Fields).

3. Machine Learning in Astronomy: Cautionary Tales for the Community
   Michelle Ntampaka.
   2022 STScI Newsletter.

2. R2—D2: Roman and Rubin — from Data to Discovery
   The AURA Roman-Rubin Synergy Working Group, including Michelle Ntampaka.
   arXiv: 2202.12311

1. Astro2020 Science White Paper: The Role of Machine Learning in the Next Decade of Cosmology
   Principal Author: Michelle Ntampaka
   Co-authors: C. Avestruz, S. Boada, J. Caldeira, J. Cisewski-Kehe, R. Di Stefano, C. Dvorkin, A. Evrard, A. Farahi, D. Finkbeiner, S. Genel, A. Goodman, A. Goulding, S. Ho, A. Kosowsky, P. La Plante, F. Lanusse, M. Lochner, R. Mandelbaum, D. Nagai, J. Newman, B. Nord, J.E.G. Peek, A. Peel, B. Póczos, M.M. Rau, A. Siemiginowska, D.J. Sutherland, H. Trac, B. Wandelt.
   Astro2020: Decadal Survey on Astronomy and Astrophysics, science white papers, no. 14.
   2019 Bulletin of the AAS, 51, 3.

Selected Professional Service

Science Organizing Committee ∘ Rare Gems in Big Data ∘ NOIRLab ∘ 2023

Science Organizing Committee ∘ Artificial Intelligence for Science and Operations in Astronomy ∘ ESA/ESO SciOps Workshop ∘ 2022

Science Organizing Committee ∘ Machine Learning Tools for Research in Astronomy ∘ Max Planck Society Schloss Ringberg ∘ 2019

Mentor ∘ The Supernova Foundation: Mentoring for Women in Physics ∘ 2018

- Mentored two women from Africa who are pursuing higher education in physics.

Astro 2020 Decadal Survey on Astronomy and Astrophysics ∘ Early Career Focus Session ∘ Keck Center of the National Academies of Sciences, Engineering, and Medicine ∘ 2018

Strategic Planning Committee ∘ CMU ∘ 2017

- Represented graduate student interests as the student representative at departmental strategic planning committee meetings.

Constructive Interference: Women and Minorities in Physics ∘ **Founder** ∘ CMU ∘ 2017

- Organized and led a biweekly reading and discussion group of physics department members.
- This thriving group is credited with attracting an unprecedentedly diverse graduate student group.

Local Organizing Committee, Statistical Challenges in Modern Astronomy VI ∘ CMU ∘ 2016

- Provided logistical support to the Scientific Organizing Committee of this international conference held at Carnegie Mellon.

University Childcare Committee ∘ CMU ∘ 2015-2016

- Collaborated with representatives from across the campus community to establish a new university child care center, making quality child care more accessible to students, staff, and faculty.

Choosing an Advisor Panel ∘ First Year Student Colloquium ∘ CMU ∘ November 2015

- Discussed topics for consideration when choosing a research advisor with junior graduate students.

Referee/Reviewer for:

- American Statistical Association Astrostatistics Interest Group Student Paper Competition
- Astronomy & Astrophysics
- Astrophysical Journal
- Canada Research Chairs Program
- *Chandra* Peer Review (Panel Deputy Chair)
- Harvard Data Science Review
- Monthly Notices of the Royal Astronomical Society
- NASA Astrophysics Data Analysis Program (Panel Chair)
- NSF Astronomy & Astrophysics Grants
- NeurIPS
- U.S.-Israel Binational Science Foundation

STScI Functional Contributions

AI Community of Practice ∘ 2025

- Developed and led a framework for the AI Community of Practice lessons and meetings.

AI Task Force ∘ 2025

- Drafted core guidelines for the responsible use of AI and worked in partnership with the other members of the AI Task Force to develop the draft into Institute policy.
- Composed a comprehensive AI Strategy after interviewing key stakeholders across the institute, including chief engineers and members of the leadership team.
- Presented the AI Strategy to the director's office, the STIC, and at an STScI All-Hands.

Vela NSF AI Institute Proposal ∘ 2023-2024

- Led the development of the proposal document by managing the collaborative writing of the $20 million strategic proposal to establish a Center for Astronomical Data Science at STScI.
- Served as the overall second-in-command for the proposed work, which included managing a cross-disciplinary, cross-institute team and making decisions about resource allocation at the multi-million-dollar level.
- Proposal Outcome: The pre-proposal, submitted October 2023, was highly rated and the NSF review panel recommended that a full proposal be submitted. The full proposal was approved by the DO and the STIC, and was highly ranked (though not ultimately funded) by the NSF.

MAST/Roman Upscope Proposal ∘ 2023

- Led a key iteration of this upscope proposal in 2023 (one year before it was ultimately accepted and funded) by building trust between MAST and PSO and collaboratively crafting a persuasive case for an additional $10 million in funding to Roman.

Amazon Web Services (AWS) Open Data Plan (ODP) Allocation Policy ∘ 2023-2024

- Coordinated stakeholders to develop and author a strategy to manage the AWS S3 resource, valued at $30 million.

Strategic Planning Committee ∘ 2022-present (work ongoing)

- Co-led the development of the strategic plank to *Establish STScI as a Leader In Astronomical Data Science*.

Pyxis Initiative ∘ 2022-2023

- Consulted with INS leadership to develop projects that explore the use of machine learning in the cloud. This program has dual goals of 1) creating products that can be assessed for deployment in AWS for *Hubble* operations and 2) growing expertise with AWS among INS staff.

Hello Universe ∘ 2022-present (work ongoing)

- Initiated an effort to develop MAST high-level science products (HLSPs) paired with Jupyter notebooks that apply ML algorithms to these data. Since their release in Summer 2022, these have been available to the astronomical community for teaching and learning machine learning with astronomical data, and also for benchmarking the performance of new ML algorithms. My role in this program was to provide the scope and vision for the work and also to build the prototype notebook. A member of the MAST team developed the published notebooks and is currently supporting community contributions to this resource.

AWS Exploratory Accounts ∘ 2022-present (work ongoing)

- Led the development of this program to give STScI researchers access to commercial cloud resources for scientific exploration.

Machine Learning Support ∘ 2020-2022

- ∘ Provided support and feedback to researchers pursuing ML-focused projects at organized ML Drop-In Hours and also during individual consultations. The projects discussed ranged from ML Operations (e.g., for WFC3 contaminant detection and library bibliographic automation) to more science-focused applications of ML.
- ∘ Taught an introductory-level deep learning course for professional engineers and astronomers.

Other Contributions:

- ∘ Led the DSMO response to NASA RFI on Scientific Data and Computing Architecture to Support Open Science to inform SMD policy SPD 41a (2023).
- ∘ Coordinated the DSMO response to the data- and data-science-focused aspects of the Astro2020 decadal (2022).
- ∘ Interviewed candidates for several roles, including: DSMO Head (2020), DSMO Project Scientist (2021), SRC (2021), DSMO Mission Scientist (2022), Roman MO Head (2023), DSMO Mission Scientist (2023), SCOPE Division Head (2023), ITSD Deputy Head (2023).
- ∘ Contributed to the DOE-NASA RFI regarding Roman-Rubin-Euclid synergies (2021).

STScI Committees and Service

Strategic DRF Committee ∘ 2026

Roman Mission Head Hiring Committee ∘ 2023

AURA Roman-Rubin Synergy Working Group ∘ 2021-2022

- ∘ Investigated which general astrophysics areas would benefit most from Roman-Rubin joint analysis, what science questions could be uniquely be addressed by Roman-Rubin working together, and how the coordination of observations, data analysis and data archiving could best realize these synergies.

Science Organizing Committee ∘ Galaxy Clusters 2022: Challenging Our Cosmological Perspectives ∘ STScI Spring Symposium ∘ 2022

Research Science Allocation (DDRF) Committee ∘ 2021-present

Research Computing Forum ∘ 2020-2022

- ∘ Contributed to ongoing support to address science staff computing needs.

Roman Computing and Data Resource Allocation Working Group ∘ 2020-2021

- ∘ Contributed to the development of a preliminary plan for Roman data management.

Lasker Postdoctoral Fellowship Committee Chair ∘ 2020

- ∘ Oversaw the committee to evaluate the applications and rank the top candidates for the 2021 Lasker Postdoctoral Fellow.
- ∘ Developed a grading rubric to fairly assess applications of postdoctoral fellowship candidates. This rubric has been adapted for use by the NASA Hubble Fellowship Program. In an evaluation of the new grading rubric, a member of the NASA Hubble Fellowship Committee said "The impression of the leads was that the rubric greatly helped to focus the discussion, brought the question of leadership to greater prominence, and the Golden Buzzer [a system for highlighting qualifications that were not accounted for in the other categories], in particular, helped highlight issues related to diversity. The feedback we received from the panelists on the rubric was quite positive."

Selected Invited Talks

Statistical Methods in Fundamental Physics Data Analyses ∘ Banff Int'l Research Station ∘ 2026
MPIfR ∘ 2025
LMU ∘ 2025
Cosmic Cartography with Roman ∘ STScI ∘ 2025
University of Chicago ∘ 2025
Massachusetts Institute for Technology ∘ 2025
Texas A&M ∘ 2025
Harvard ∘ Institute for Theory and Computation Lunch ∘ 2024
Center for Astrophysics | Harvard & Smithsonian ∘ Galaxy Cluster Group Meeting ∘ 2024
AstroAI Asian Network (A^3Net) Summer School ∘ Osaka, Japan ∘ 2024
Kavli Institute for the Physics and Mathematics of the Universe (IPMU) ∘ Tokyo, Japan ∘ 2024
Invited Keynote ∘ NSF Postdoc Symposium ∘ 2024
AAS 245th Meeting ∘ The Impact of Artificial Intelligence on Astrophysics Research ∘ 2024
Astroinformatics ∘ Naples, Italy ∘ 2023
Aspen Center for Physics ∘ Workshop on Galaxy Clusters ∘ 2023
Princeton Career Panel ∘ 2022
Astro Hack Week ∘ Germany ∘ 2022
Cosmology from Home ∘ 2022
European Astronomical Society Annual Meeting ∘ Valencia, Spain ∘ 2022
ESA/ESO SciOps: Artificial Intelligence for Science and Operations in Astronomy ∘ 2022
STScI Spring Symposium ∘ Galaxy Clusters: Challenging Our Cosmological Perspectives ∘ 2022
Université de Montréal ∘ 2022
A Multi-Wavelength View of Galaxy Clusters ∘ ESA ∘ 2021
Fermilab Cosmic Physics Center ∘ 2021
Chandra Data Science: Novel Methods in Computing and Stats for X-ray Astronomy ∘ 2021
ESA Science Directorate ∘ 2021
ΣΠΣ Induction ∘ Washington & Jefferson College ∘ 2021
Physics $\bigcap$ ML ∘ 2021
AAS 237th Meeting ∘ Education in Astrostatistics and Astroinformatics ∘ 2021
Pennsylvania State University ∘ 2020
University of California Berkeley ∘ 2020
Max-Planck-Institut für Astrophysik ∘ 2019
Michigan Institute for Data Science (MIDAS) Symposium ∘ 2019
University of Texas Austin ∘ 2019
University of Illinois at Urbana-Champaign ∘ 2019
University of Michigan ∘ 2019
Forum on the Impact of Big Data in Astronomy ∘ International Space Science Institute ∘ Bern, Switzerland ∘ 2019
Artificial Intelligence Methods in Cosmology ∘ ETH Zürich ∘ 2019
University of Chicago ∘ 2019
Massachusetts Institute of Technology ∘ 2017

Selected Contributed Talks

Aspen Center for Physics Summer Workshop ∘ 2023

AAS 240th Meeting ∘ 2022

Astronomical Data Analysis Software and Systems (ADASS) XXXI ∘ University of Cape Town ∘ 2021

Machine Learning Tools for Research in Astronomy ∘ Max-Planck-Institut ∘ 2019

lllustrisTNG Science Meeting ∘ Max-Planck-Institut für Astrophysik ∘ 2018

Machine Learning in Science and Engineering ∘ Carnegie Mellon University ∘ 2018

Yale University ∘ 2018

AAS 231st Meeting ∘ Applied Statistical Methods in Astronomy: Gaussian Processes and Machine Learning ∘ 2018

Kavli Institute for Astrophysics and Space Research ∘ MIT ∘ 2017

Statistical Challenges in Modern Astronomy VI ∘ Carnegie Mellon University ∘ 2016

Neighborhood Workshop on Astrophysics and Cosmology ∘ Pennsylvania State University ∘ 2016

SnowPAC ∘ University of Utah ∘ 2016

AAS 227th Meeting ∘ 2016

Astronomy Ambassadors Alumni Panel
AAS 227th Meeting ∘ Kissimmee, Florida ∘ 2016

Innovative Simulations with Machine Learning & Statistics in the Era of LSST ∘ CMU ∘ 2015

Neighborhood Workshop on Astrophysics and Cosmology ∘ Pennsylvania State University ∘ 2015

- ∘ **Awarded Best Student Talk.**

Advising Experience

Postdoctoral Researchers:

Ana Maria Delgado ∘ Johns Hopkins University ∘ 2024-2026

- ∘ **2026 Faculty at The City University of New York (CUNY)**

Graduate Students:

John Soltis ∘ Johns Hopkins University ∘ Ph.D. Advisor ∘ 2021-2025

- ∘ Thesis: A Multi-Wavelength View of Galaxy Cluster Mass Accretion
- ∘ **Graduated 2025; Lasker Prize Postdoctoral Fellowship.**

Brianna Galgano ∘ Johns Hopkins University ∘ Research Rotation Advisor ∘ 2020-2023

- ∘ Project: A Deep Learning Cluster Finder for eROSITA

Matthew Ho ∘ Carnegie Mellon University ∘ Paper Co-Advisor ∘ 2021

- ∘ Paper title: Deep Learning and the Dynamical Mass of the Coma Cluster

V. Ashley Villar ∘ Harvard University ∘ Paper Co-Advisor ∘ 2019-2020

- ∘ Paper title: SuperRAENN: A Semisupervised Supernova Photometric Classification Pipeline Trained on Pan-STARRS1 Medium-Deep Survey Supernovae

Sheridan Green ∘ Yale University ∘ Paper Co-Advisor ∘ 2019

- ∘ Paper title: Using X-Ray Morphological Parameters to Strengthen Galaxy Cluster Mass Estimates via Machine Learning

Undergraduate Students:

Suchitra Narayanan ∘ The Banneker Institute ∘ Harvard University ∘ REU Advisor ∘ 2019

- ∘ Project title: Galaxy Cluster Membership with Machine Learning
- ∘ Student received an honorable mention in the AAS Chambliss undergraduate poster competition.

Nicel Mohamed-Hinds ∘ The Banneker Institute ∘ Harvard University ∘ REU Advisor ∘ 2018

- ∘ Project title: Galaxy Cluster Mass Estimates With a Random Forests Regressor
- ∘ Student received an honorable mention in the AAS Chambliss undergraduate poster competition.

Committee Member for Master's and PhD Theses:

Verlon Etsebeth ∘ University of the Western Cape ∘ South Africa ∘ 2025
Koketso Mohale ∘ University of the Western Cape ∘ South Africa ∘ 2024
Carter Rhea ∘ Université de Montréal ∘ Montreal, Canada ∘ 2024

Classroom Teaching Experience

Special Lecturer & Laboratory Technician ∘ CMU ∘ 2005-2012

Experimental Physics (33-104) ∘ Course Instructor ∘ 2005-2012

Light from the Enlightenment (99-243) ∘ Interdisciplinary Course ∘ 2011

- ∘ Course co-creator and co-instructor.
- ∘ Course description: an introduction to classical optics, from historical origins to contemporary applications. The course emphasizes experimental observations guided by primary scientific texts, initially focused on the Enlightenment (e.g., Galileo, Newton, Huygens) and moving towards current theories. Additionally, the course provides a contextual understanding of how the comprehension and manipulation of light have affected the way we see ourselves through art, technology, and society.

Physics Teacher ∘ Baldwin High School ∘ 2004-2005

Conceptual Physics and Academic Physics

- ∘ Developed and implemented a dynamic curriculum for a new conceptual physics course.
- ∘ Guided students to understanding through the use of hands-on activities and labs, engaging demonstrations, and classroom discussion.

Science Teacher ∘ Grove City Academy ∘ 2003-2004

Biology (9th grade) & Earth and Space Sciences (6th grade)

- ∘ Implemented best teaching practices to engage students with lessons and activities that were adapted for multiple intelligences (e.g. lessons for visual learners, kinesthetic learners, and auditory learners).

Educational Outreach and Science Communication

Interviewed for NPR's Morning Edition ∘ April 2019

Moonbear's Shadow ∘ Cunniff Elementary School ∘ September 2018

- ∘ Explored light and shadow with kindergarten classes through story and play.

Solar Observing ∘ Poff Elementary School ∘ May 2017

- ∘ Demonstrated safe solar observing for the kindergarten class and their families, with a focus on how to safely observe the 2017 solar eclipse.

Public Lecture: Using Machine Learning to Understand the Makeup of the Universe
The Allegheny Observatory Public Lecture ∘ Pittsburgh, PA ∘ October 2016

Meet the Scientists ∘ Carnegie Science Center ∘ May 2015

- ∘ Engaged museum visitors with a hands-on activity, using it as a springboard to discuss dark matter, galaxy clusters, and how my research relates to both.

Rwanda Solar Eclipse Outreach ∘ November 2013

- ∘ Trained a core group of science teachers to explain the partial solar eclipse, and together performed public outreach in conjunction with this event.

Storyteller ∘ Under the African Stars ∘ The Story Collider ∘ October 2013

- ∘ Developed and performed a personal story about stargazing in Africa at a live storytelling event.

Astronomy Ambassador ∘ AAS ∘ 2013

- ∘ Created outreach materials using best teaching practices as a member of the inaugural class of AAS Ambassadors.

Transit of Venus ∘ Organizing Committee Member ∘ CMU ∘ June 2012

- ∘ Constructed stations for safe solar observing and interacted with astronomy enthusiasts who came to campus to view this rare astronomical event.

Science Writing ∘ CMU English Department ∘ Fall Semester 2011

- ∘ Developed a newspaper-style science article portfolio over this semester-long course. In an evaluation, the course instructor wrote: "She has a great gift for taking what she knows, and what to most would seem abstract and mysterious, and shining a light on it so that even humble science writers can grasp its logic and importance."

Rwanda Science and Mathematics Textbook Drive ∘ 2011

- ∘ Oversaw the collection of approximately 4,500 science and mathematics textbooks, organized their delivery to Kigali, Rwanda, and directed their distribution by the Rwanda Ministry of Education to supply hundreds of Rwandan secondary science and mathematics teachers with reference textbooks.

Rwanda Teacher Training ∘ Lycée de Kigali ∘ Kigali, Rwanda ∘ June 2011

- ∘ Initiated contact with the Rwandan Minister of Education to establish a teacher training seminar. I developed and taught a curriculum to teach toward three goals: using commonly available items to teach through lecture demonstrations, maintaining students' attention through long lessons, and efficiently planning effective lessons.

Professional Certifications

Physics Teaching Certification ∘ Pennsylvania ∘ 2004-present

Physics: Received ETS Recognition of Excellence Award ∘ Physics PRAXIS ∘ 2004

General Science Secondary Teaching Certification ∘ Pennsylvania ∘ 2004-present